

UNIT 6

❖ ELEMENTARY ROW TRANSFORMATION/OPERATION ON A MATRIX:

- ✓ R_{ij} or $R_i \leftrightarrow R_j$ means interchange of i^{th} and j^{th} rows.
- ✓ $k \cdot R_i$ means multiplication of all the elements of i^{th} row by non-zero scalar k .
- ✓ $R_{ij}(k)$ or $R_j + k \cdot R_i$ means multiplication of all the elements of i^{th} row by nonzero scalar k and add it into j^{th} row.

❖ ROW ECHELON FORM OF MATRIX:

- ✓ Procedure to convert the matrix into Row Echelon Form is as follow:
 - (1). Every zero row of the matrix occurs below the non-zero rows.
 - (2). Arrange all the rows in decreasing order.
 - (3). Make all the entries zero below the leading (first non-zero entry of the row) element of 1^{st} row.
 - (4). Repeat step-3 for each row.

❖ REDUCED ROW ECHELON FORM OF MATRIX:

- ✓ Procedure to convert the matrix into Reduced Row Echelon Form is as follow:
 - (1). Convert the given matrix into Row Echelon Form.
 - (2). Make all the leading elements 1(one).
 - (3). Make all the entries zero above the leading element 1(one) of each row.

❖ RANK OF A MATRIX:

- ✓ Let A be an $m \times n$ matrix. Then the positive integer r is said to be the rank of A if
 - (1). There is at least one minor of order r which is non-zero.
 - (2). Every $(r + 1)^{\text{th}}$ order minor of A is zero.
- ✓ Rank of matrix A is denoted by $\rho(A)$ **OR** $\text{Rank}(A)$.

❖ NOTES:

- ✓ $\rho(AB) \leq \rho(A)$ **OR** $\rho(AB) \leq \rho(B)$.

- ✓ $\rho(A) = \rho(A^T)$.
- ✓ For a matrix A of order $m \times n$, $\rho(A) \leq \min\{m, n\}$.
- ✓ Procedure to find the Rank of given matrix is as follow:
 - (1). Reduced the given matrix in Row Echelon Form.
 - (2). Number of non-zero row in Row-Echelon Form is the Rank of given matrix.

METHOD – 1: ECHELON FORM AND RANK OF MATRIX

C	1	Determine whether the following matrices are in Row-Echelon Form, Reduced Row-Echelon Form, both or none. (1). $\begin{bmatrix} 3 \end{bmatrix}$ (2). $\begin{bmatrix} 1 & 2 \end{bmatrix}$ (3). $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (4). $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (5). $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix}$ (6). $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (7). $\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}$ (8). $\begin{bmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}$ (9). $\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ (10). $\begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$. Answer: (1). Echelon Form (2). & (3). Both (4). Echelon Form (5). None (6). None (7). Both (8). & (9). Echelon Form (10). Both.	
H	2	Is the matrix $\begin{bmatrix} 1 & 0 & 4 & 6 \\ 0 & 1 & 5 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ in Row Echelon Form or Reduced Row Echelon Form? Answer: Echelon Form.	
C	3	Find Reduced Row Echelon Form of $A = \begin{bmatrix} 2 & 4 & 3 & 4 \\ 1 & 2 & 1 & 3 \\ 1 & 3 & 2 & 2 \\ 3 & 7 & 4 & 8 \end{bmatrix}$. Answer: $\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.	

H	4	Find Reduced Row Echelon Form of $A = \begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$.	
		Answer: $\begin{bmatrix} 1 & 2 & 0 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}$.	
T	5	Convert $\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{bmatrix}$ into its equivalent Reduced Row Echelon Form.	
		Answer: $\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.	
C	6	Reduce matrix $A = \begin{bmatrix} 1 & 5 & 3 & -2 \\ 2 & 0 & 4 & 1 \\ 4 & 8 & 9 & -1 \end{bmatrix}$ to Row Echelon Form and find its Rank.	W-18 (3)
		Answer: $\begin{bmatrix} 1 & 5 & 3 & -2 \\ 0 & -10 & -2 & 5 \\ 0 & 0 & -3/5 & 1 \end{bmatrix}$ & $\rho(A) = 3$.	
H	7	Find the Rank of the following matrices: $\begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 4 & 8 & 12 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 4 & 8 & 12 & 16 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 3 & -1 \\ -2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & -7 \end{bmatrix}$, $\begin{bmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 & 2 & 0 \\ 3 & 1 & 0 & 0 \\ -1 & 2 & 4 & 0 \end{bmatrix}$ & $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$. Answer: 1, 1, 2, 2, 2, 3, 3 & 3.	
T	8	Define the rank of a matrix and find the rank of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix}$. Answer: $\rho(A) = 3$.	S-19 (3)

C	9	Convert the following matrix in to Reduced Row Echelon Form and hence find the Rank of a matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$. Answer: $\begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix}$ & $\rho(A) = 2$.	
H	10	Find the Rank of the matrix $A = \begin{bmatrix} 1 & 2 & 4 & 0 \\ -3 & 1 & 5 & 2 \\ -2 & 3 & 9 & 2 \end{bmatrix}$. Answer: $\rho(A) = 2$.	
H	11	Find the Rank of the matrix $A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$. Answer: $\rho(A) = 2$.	
H	12	Find the Rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Answer: $\rho(A) = 2$.	
T	13	Find the Rank of the matrix $A = \begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix}$. Answer: $\rho(A) = 2$.	
C	14	Find the Rank of the matrix $A = \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 9 \\ 10 & 11 & 12 & 13 & 14 \\ 15 & 16 & 17 & 18 & 19 \end{bmatrix}$. Answer: $\rho(A) = 2$.	
H	15	Obtain the Reduced Row Echelon Form of the matrix $A = \begin{bmatrix} 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 8 \end{bmatrix}$ and hence find the Rank of the matrix A. Answer: $\rho(A) = 3$.	
C	16	Find the Rank of $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{bmatrix}$ in terms of determinants. Answer: $\rho(A) = 2$.	

- ✓ In Gauss Elimination and Gauss-Jordan method if $\rho(A) = \rho(A : B)$ then system is consistent otherwise it is inconsistent.

❖ HOMOGENEOUS AND NON-HOMOGENEOUS SYSTEM

- ✓ A linear system $AX = B$ is called **Homogeneous** linear system if matrix $B = 0$ (null or zero column matrix). Otherwise it is called **Non-homogeneous** linear system.

- ✓ Non-homogeneous linear system (i. e. $B \neq 0$) has three types of solutions.

(1). No solution, (2). Unique solution & (3). Infinitely many solutions.

- ✓ Homogeneous linear system (i. e. $B = 0$) has two types of solutions.

(1). Unique solution (trivial solution: $(0, 0, \dots, 0)$)

(2). Infinitely many solutions (non-trivial solution)

- ✓ For the linear system in which no. of equations = no. of unknown (square system),

No solution $\Rightarrow [0 \ 0 \ \dots \ 0 : \#]$

Unique solution $\Rightarrow [0 \ 0 \ \dots \ \# : 0]$ or $[0 \ 0 \ \dots \ \# : *]$

Infinite solutions $\Rightarrow [0 \ 0 \ \dots \ 0 : 0]$

- ✓ Solution by rank:

$\rho(A) = \rho(A : B) = n \rightarrow$ Unique solution,

$\rho(A) = \rho(A : B) < n \rightarrow$ Infinitely many solution,

$\rho(A) \neq \rho(A : B) \rightarrow$ No solution, where $n =$ no. of unknowns.

❖ NOTES:

- ✓ In the case of infinitely many solutions:

➤ The number of free variables = $n - \rho(A)$, where $n =$ no. of unknowns.

- ✓ If the Non-Homogeneous system of equations $AX = B$ is square system (no. of unknown = no. of equations) then find the determinant of co-efficient matrix A .

➤ If $\det(A) \neq 0$, then the system of linear equations has unique solution.

➤ If $\det(A) = 0$, then the system of linear equations has either no solution or infinitely many solutions.

- ✓ If the homogeneous system of equations $AX = \mathbf{0}$ is square system (no. of unknown = no. of equations) then find the determinant of co-efficient matrix A.
 - If $\det(A) \neq 0$, then the system of linear equations has unique solution It means trivial solution.
 - If $\det(A) = 0$, then the system of linear equations has infinitely many solutions.

❖ **NOTE:**

- ✓ Since the homogeneous system $AX = \mathbf{0}$ contains trivial solution, this system is always consistent.

❖ **GAUSS ELIMINATION METHOD TO SOLVE SYSTEM OF LINEAR EQUATIONS:**

- ✓ Procedure to solve the given system of linear equations using Gauss Elimination Method:
 - (1). Start with augmented matrix $[A : B]$.
 - (2). Convert $[A : B]$ into Row Echelon Form with leading element of each row is one (1).
 - (3). Apply back substitution for getting equations.
 - (4). Solve the equations and find the unknown variables (i.e. solution).

METHOD - 2: GAUSS ELIMINATION METHOD

C	1	Using Gauss Elimination method solve the following system $-x + 3y + 4z = 30$, $3x + 2y - z = 9$, $2x - y + 2z = 10$. Answer: (2, 4, 5).	W-19 (7)
H	2	Solve by using Gauss Elimination Method: (1). $x_1 + x_2 + 2x_3 = 8$, $-x_1 - 2x_2 + 3x_3 = 1$, $3x_1 - 7x_2 + 4x_3 = 10$. (2). $x_1 + 2x_2 + 3x_3 = 4$, $2x_1 + 5x_2 + 3x_3 = 5$, $x_1 + 8x_3 = 9$. Answer: (1). (3, 1, 2) & (2). (1, 0, 1).	
H	3	Solve the given system of linear equations by using Gauss Elimination Method: $2x + y - z = 4$, $x - y + 2z = -2$, $-x + 2y - z = 2$. Answer: (1, 1, -1).	

H	4	Solve system of linear equations by using Gauss Elimination method, if solution exists. $x + y + 2z = 9$, $2x + 4y - 3z = 1$, $3x + 6y - 5z = 0$. Answer: (1, 2, 3).	W-18 (4)
C	5	Solve by using Gauss Elimination Method: $-\frac{1}{x} + \frac{3}{y} + \frac{4}{z} = 30$, $\frac{3}{x} + \frac{2}{y} - \frac{1}{z} = 9$, $\frac{2}{x} - \frac{1}{y} + \frac{2}{z} = 10$. Answer: (1/2, 1/4, 1/5).	
T	6	Solve the given system of linear equations by using Gauss Elimination Method: $x + y + z = 3$, $x + 2y - z = 4$, $x + 3y + 2z = 4$. Answer: (13/5, 3/5, -1/5).	
T	7	Solve the given system of linear equations using Gauss Elimination Method: $2x_1 + x_2 + 2x_3 + x_4 = 6$, $6x_1 - x_2 + 6x_3 + 12x_4 = 36$, $4x_1 + 3x_2 + 3x_3 - 3x_4 = 1$, $2x_1 + 2x_2 - x_3 + x_4 = 10$. Answer: (7/4, 3/2, -5/6, 8/3).	
C	8	Solve by using Gauss Elimination Method: $-2b + 3c = 1$, $3a + 6b - 3c = -2$, $6a + 6b + 3c = 5$. Answer: No solution.	
H	9	Solve by using Gauss Elimination Method: (1). $-2y + 3z = 1$, $3x + 6y - 3z = -2$, $6x + 6y + 3z = 5$. (2). $3x + y - 3z = 13$, $2x - 3y + 7z = 5$, $2x + 19y - 47z = 32$. Answer: No solution for each system.	
C	10	Solve the given system of linear equations by using Gauss Elimination Method: $x_1 - 2x_2 + 3x_3 = -2$, $-x_1 + x_2 - 2x_3 = 3$, $2x_1 - x_2 + 3x_3 = -7$. Answer: $\{(-4 - t, t - 1, t)/t \in \mathbb{R}\}$.	

C	11	Solve by using Gauss Elimination Method: $x_1 + 3x_2 - 2x_3 + 2x_5 = 0, \quad 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1,$ $5x_3 + 10x_4 + 15x_6 = 5, \quad 2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6.$ Answer: $x_1 = -3r - 4s - 2t$, $x_2 = r$, $x_3 = -2s$, $x_4 = s$, $x_5 = t$, $x_6 = 1/3$; where $r, s, t \in \mathbb{R}$.	
H	12	Solve by using Gauss Elimination Method: $x_1 - 2x_2 - x_3 + 3x_4 = 1, \quad 2x_1 - 4x_2 + x_3 = 5, \quad x_1 - 2x_2 + 2x_3 - 3x_4 = 4.$ Answer: $x_1 = 2 + 2t_2 - t_1$, $x_2 = t_2$, $x_3 = 1 + 2t_1$, $x_4 = t_1$, $t_1, t_2 \in \mathbb{R}$.	
T	13	Solve by using Gauss Elimination Method: (1). $5x + 3y + 7z = 4$, $3x + 26y + 2z = 9$, $7x + 2y + 10z = 5$. (2). $2x_1 + 2x_2 + 2x_3 = 0$, $-2x_1 + 5x_2 + 2x_3 = 1$, $8x_1 + x_2 + 4x_3 = -1$. Answer: (1). $\{((7 - 16t)/11, (3 + t)/11, t)/t \in \mathbb{R}\}$ & (2). $\{x_1 = (-1 - 3t)/7, x_2 = (1 - 4t)/7, x_3 = t; \text{ where } t \in \mathbb{R}\}$.	
C	14	The augmented matrix of a linear system has the form $\begin{bmatrix} -2 & 3 & 1 & : & a \\ 1 & 1 & -1 & : & b \\ 0 & 5 & -1 & : & c \end{bmatrix}$. Determine when the linear system is consistent. Determine when the linear system is inconsistent. Does the linear system have a unique solution or infinitely many solutions? Answer: $a + 2b - c = 0$, $a + 2b - c \neq 0$ & Infinitely many solutions.	
H	15	Determine when the given augmented matrix represents a consistent linear system. $\begin{bmatrix} 1 & 0 & 2 & : & a \\ 2 & 1 & 5 & : & b \\ 1 & -1 & 1 & : & c \end{bmatrix}$ Answer: $b + c - 3a = 0$.	
H	16	For what choices of parameter λ the following system is consistent? $x_1 + x_2 + 2x_3 + x_4 = 1, \quad x_1 + 2x_3 = 0, \quad 2x_1 + 2x_2 + 3x_3 = \lambda,$ $x_2 + x_3 + 3x_4 = 2\lambda.$ Answer: For $\lambda = 1$ given system is consistent.	

T	17	Determine the values of k , for which the equations $3x - y + 2z = 1$, $-4x + 2y - 3z = k$, $2x + z = k^2$ possesses solution. Find solutions in each case. Answer: $k = 2 \Rightarrow \{(4 - t)/2, (10 + t)/2, t\} / t \in \mathbb{R}$ & $k = -1 \Rightarrow \{(1 - t)/2, (1 + t)/2, t\} / t \in \mathbb{R}$.	
C	18	For which value of λ and k the following system has (i) No solution, (ii) Unique solution, (iii) An infinite no. of solution. $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = k$. Answer: (i) $\lambda = 3$, $k \neq 10 \Rightarrow$ no solution, (ii) $\lambda \neq 3$, $k \in \mathbb{R} \Rightarrow$ unique solution & (iii) $\lambda = 3$, $k = 10 \Rightarrow$ infinite solutions.	
H	19	For which values of 'a' will the following system has (i) no solution, (ii) unique solution, (iii) Infinitely many solutions. $x + 2y - 3z = 4$, $3x - y + 5z = 2$, $4x + y + (a^2 - 14)z = a + 2$. Answer: (i) $a = -4 \Rightarrow$ no solution, (ii) $a \in \mathbb{R} - \{-4, 4\} \Rightarrow$ unique solution & (iii) $a = 4 \Rightarrow$ infinite solutions.	
T	20	Determine the value of k so that the system of homogeneous equations $2x + y + 2z = 0$, $x + y + 3z = 0$, $4x + 3y + kz = 0$ has (a) Trivial solution. (b) Non-trivial solution. Also find non-trivial solution. Answer: (a) If $k \neq 8 \Rightarrow$ trivial solution. i.e. $(0, 0, 0)$ & (b) If $k = 8 \Rightarrow x = k; y = -4k; z = k$, where k is a parameter.	
C	21	Find real value of λ for which the equations $(1 - \lambda)x - y + z = 0$, $2x + (1 - \lambda)y = 0$, $2y - (1 + \lambda)z = 0$ have solution other than $x = y = z = 0$. Also find the solution for each value of λ . Answer: $\lambda = 1 \Rightarrow x = 0; y = k; z = k$, where $k \in \mathbb{R}$ is arbitrary.	

❖ GAUSS-JORDAN ELIMINATION METHOD TO SOLVE THE SYSTEM OF LINEAR EQUATIONS:

✓ Procedure to solve the system of linear equations by Gauss-Jordan Elimination Method:

(1). Start with the augmented matrix $[A : B]$.

- (2). Convert $[A : B]$ into reduced row echelon form.
- (3). Find the unknowns, which is required solutions.

METHOD – 3: GAUSS - JORDAN METHOD

C	1	Solve the following system of linear equations by Gauss-Jordan Method. $x + y + 2z = 8, \quad -x - 2y + 3z = 1, \quad 3x - 7y + 4z = 10.$ Answer: (3, 1, 2).	
H	2	Solve by using Gauss-Jordan Method: (1). $x + y + z = 6, \quad x + 2y + 3z = 14, \quad 2x + 4y + 7z = 30.$ (2). $x_1 + 2x_2 + 3x_3 = 1, \quad 2x_1 + 5x_2 + 3x_3 = 6, \quad x_1 + 8x_3 = -6.$ (3). $x_1 + 2x_2 + 3x_3 = 4, \quad 2x_1 + 5x_2 + 3x_3 = 5, \quad x_1 + 8x_3 = 9.$ Answer: (1). (0, 4, 2) (2). (2, 1, -1) (3). (1, 0, 1).	
H	3	Solve the system of linear equations by using Gauss-Jordan Elimination Method. $x + 4y - 3z = 0, \quad -x - 3y + 5z = -3, \quad 2x + 8y - 5z = 1.$ Answer: (23, -5, 1).	
H	4	Solve by using Gauss-Jordan Method: (1). $2x - y - 3z = 0, \quad -x + 2y - 3z = 0, \quad x + y + 4z = 0.$ (2). $2x_1 + x_2 + 3x_3 = 0, \quad x_1 + 2x_2 = 0, \quad x_2 + x_3 = 0.$ Answer: Trivial solution (0, 0, 0) for both system.	
T	5	Solve by using Gauss-Jordan Method, where $0 \leq \alpha \leq 2\pi, 0 \leq \beta \leq 2\pi, 0 \leq \gamma \leq \pi.$ $2 \sin \alpha - \cos \beta + 3 \tan \gamma = 3, \quad 4 \sin \alpha + 2 \cos \beta - 2 \tan \gamma = 2,$ $6 \sin \alpha - 3 \cos \beta + \tan \gamma = 9.$ Answer: (1, -1, 0).	
C	6	Solve by using Gauss-Jordan Method: $-2y + 3z = 1, \quad 3x + 6y - 3z = -2, \quad 6x + 6y + 3z = 5.$ Answer: Inconsistent(no solution).	

C	7	Use Gauss-Jordan algorithm to solve the system of linear equations: $2x_1 + 2x_2 - x_3 + x_5 = 0, \quad -x_1 - x_2 + 2x_3 - 3x_4 + x_5 = 0,$ $x_1 + x_2 - 2x_3 - x_5 = 0, \quad x_3 + x_4 + x_5 = 0.$ Answer: $\{(-s - t, s, -t, 0, t) / t, s \in \mathbb{R}\}.$	S-19 (4)
H	8	Solve the system of linear equations $v + 3w - 2x = 0, \quad 2u + v - 4w + 3x = 0,$ $2u + 3v + 2w - x = 0, -4u - 3v + 5w - 4x = 0.$ Answer: $u = (7s - 5t)/2, v = 2t - 3s, w = s, x = t, \text{ where } s, t \in \mathbb{R}.$	
H	9	Solve $x_1 + x_2 + 2x_3 - 5x_4 = 3, \quad 2x_1 + 5x_2 - x_3 - 9x_4 = -3,$ $2x_1 + x_2 - x_3 + 3x_4 = -11, x_1 - 3x_2 + 2x_3 + 7x_4 = -5$ using Gauss-Jordan Method. Answer: $\{(-5 - 2t, 2 + 3t, 3 + 2t, t) / t \in \mathbb{R}\}.$	
T	10	Solve by using Gauss-Jordan Method: (1). $x_1 + 3x_2 + x_4 = 0, \quad x_1 + 4x_2 + 2x_3 = 0, \quad -2x_2 - 2x_3 - x_4 = 0.$ (2). $2x_1 - 4x_2 + x_3 + x_4 = 0, \quad x_1 - 2x_2 - x_3 + x_4 = 0.$ Answer: (1). $\{(5t, -2t, 3t/2, t) / t \in \mathbb{R}\},$ (2). $\{((6t_2 - 2t_1)/3, t_2, t_1/3, t_1) / t_1 \& t_2 \in \mathbb{R}\}.$	
T	11	Solve using by Gauss-Jordan Method: (i). $3x - y - z = 0, \quad x + y + 2z = 0, \quad 5x + y + 3z = 0 \quad \&$ (ii). $x + y - z + w = 0, \quad x - y + 2z - w = 0, \quad 3x + y + w = 0.$ Answer: (i). $\left\{ \left(x = -\frac{1}{4}t, y = -\frac{7}{4}t, z = t, \text{ where } t \in \mathbb{R} \right) \right\},$ (ii). $\left\{ \left(x = -\frac{1}{2}t_1, y = \frac{3}{2}t_1 - t_2, z = t_1, w = t_2; t_1, t_2 \in \mathbb{R} \right) \right\}.$	

❖ INVERSE BY GAUSS - JORDAN METHOD:

✓ Procedure to find the inverse of Matrix by Gauss-Jordan Method is as follow:

- (1). Start with augmented matrix $[A : I]$, where I is the identity matrix of the same size of matrix A .
- (2). Convert $[A : I]$ into reduced row echelon form.
- (3). The inverse of the matrix A is I with respective changes. i.e. $[I : A^{-1}]$.

METHOD – 4: INVERSE BY GAUSS - JORDAN METHOD

C	1	Find the inverse of the given matrix by Gauss-Jordan Method $A = \begin{bmatrix} 1 & -2 \\ 3 & -1 \end{bmatrix}.$ Answer: $A^{-1} = \frac{1}{5} \begin{bmatrix} -1 & 2 \\ -3 & 1 \end{bmatrix}.$	
H	2	If possible then find the inverse of the given matrix by Gauss-Jordan Method $A = \begin{bmatrix} 6 & -4 \\ -3 & 2 \end{bmatrix}.$ Answer: Not possible because $\det(A) = 0$.	
C	3	Find the inverse of matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ by Gauss-Jordan Method. Answer: $A^{-1} = \frac{1}{6} \begin{bmatrix} -1 & 3 & -1 \\ -7 & 3 & 5 \\ 5 & -3 & -1 \end{bmatrix}.$	
H	4	If possible then find the inverse of matrix by Gauss-Jordan Method $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix}.$ Answer: Not possible because $\det(A) = 0$.	
H	5	Find the inverse of the given matrix by Gauss-Jordan Method $A = \begin{bmatrix} 2 & 6 & 6 \\ 2 & 7 & 6 \\ 2 & 7 & 7 \end{bmatrix}.$ Answer: $A^{-1} = \begin{bmatrix} 7/2 & 0 & -3 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$	
T	6	Find the inverse of matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ using Gauss-Jordan Method. Answer: $\begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & -1/2 \end{bmatrix}.$	
T	7	Find the inverse of matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ by Gauss-Jordan Method. Answer: $A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ -8 & 6 & -2 \\ 5 & -3 & 1 \end{bmatrix}.$	

C	8	Using Gauss – Jordan method find A^{-1} for $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$.	W-19 (7)
H	9	Find the inverse of matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ using row operations. Answer: $A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$.	
H	10	Find A^{-1} for $A = \begin{bmatrix} 0 & 1 & -1 \\ 3 & 1 & 1 \\ 1 & 2 & -1 \end{bmatrix}$, if exists. Answer: $A^{-1} = \begin{bmatrix} 3 & 1 & -2 \\ -4 & -1 & 3 \\ -5 & -1 & 3 \end{bmatrix}$.	
H	11	Find the inverse of $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ using row operations. Answer: $A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$.	
T	12	Find the inverse of following matrices by Gauss-Jordan Method $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 1 & 3 & 5 & 7 \end{bmatrix}$ & $\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 2 & 1 & 2 & 1 \\ 3 & -2 & 1 & 6 \end{bmatrix}$. Answer: $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1/3 & 1/3 & 0 & 0 \\ 0 & -1/5 & 1/5 & 0 \\ 0 & 0 & -1/7 & 1/7 \end{bmatrix}$ & $\begin{bmatrix} 2 & -1 & 1 & -1 \\ -5 & -3 & 1 & 1 \\ 2 & 3 & -1 & 0 \\ -3 & -1 & 0 & 1 \end{bmatrix}$.	
T	13	Find the inverse of $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ using Gauss-Jordan Method. Answer: $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 1 & -2 & 2 & -3 \\ 0 & 1 & -1 & 1 \\ -2 & 3 & -2 & 3 \end{bmatrix}$.	

❖ EIGENVALUES, EIGENVECTORS & EIGEN SPACE:

- ✓ Let A be an $n \times n$ matrix, then the scalar λ is called an Eigen Value of A , if there exist a nonzero vector X such that $AX = \lambda X$.
- ✓ The non-zero vector X is called an Eigen Vector of A corresponding to λ .
- ✓ The solution space of system $[A - \lambda I]X = 0$ is called Eigen Space of matrix A corresponding to λ .

❖ NOTES:

- ✓ Let A be an $n \times n$ matrix then

(1). An Eigen Value of A is a scalar λ such that $\det(A - \lambda I) = 0$.

(2). The Eigen Vectors of A corresponding to λ are the non-zero solutions of $(A - \lambda I)X = 0$. Here $\det(A - \lambda I) = 0$ is called the Characteristic Equation of A .

- ✓ Eigen value is also known as characteristic root/latent value/proper roots and Eigen Vector is also known as characteristic vector/latent vector/proper vector corresponding to the eigen value λ of the matrix.
- ✓ The set of all eigenvalues of matrix A is called Spectrum of A .
- ✓ The characteristic equation of 2×2 matrix A is $\lambda^2 - S_1\lambda + S_2 = 0$.

Where, S_1 = Sum of the principal diagonal elements (Trace(A)) and S_2 = Determinant of A .

- ✓ The characteristic equation of 3×3 matrix A is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$.

Where S_1 = Sum of the principal diagonal elements or Trace(A),

S_2 = Sum of the minors of the principal diagonal elements &

S_3 = Determinant of A .

- ✓ Procedure for finding eigen values, eigen vectors and eigen spaces for 3×3 matrix A .
 - (1). Find the characteristic equation $\det(A - \lambda I) = 0$. It will be a polynomial equation of degree 3 in the variable λ .
 - (2). Find the roots of the characteristic equation. These are the Eigen Values of A , say $\lambda_1, \lambda_2, \lambda_3$.

- (3). For each Eigen Value λ_i , find the Eigen Vectors corresponding to λ_i by solving the Homogeneous system $[A - \lambda_i I] \cdot X = 0$. (Here the solution space of this system is Eigen Space corresponding to λ_i).

✓ The procedure remains same for the matrices of order 2.

❖ **PROPERTIES OF EIGEN VALUES AND EIGEN VECTORS:**

- ✓ $\text{Trace}(A) = \text{Sum of the eigen values of matrix } A$.
- ✓ $\text{Det}(A) = \text{Product of the eigen values of matrix } A$.
- ✓ The eigen values of triangular matrix are the elements on its principal diagonal.
- ✓ If one of the eigen value is zero then $\det(A) = 0$, hence A^{-1} does not exist.
- ✓ The square matrix A and A^T have the same eigen values but eigen vectors need not to be same.
- ✓ If λ is an eigen value of a non-singular matrix A and X is an eigen vector correspond to λ then $\frac{1}{\lambda}$ is an eigen value of A^{-1} and X is an eigen vector correspond to $\frac{1}{\lambda}$.
- ✓ If λ is an eigen value of matrix A and X is an eigen vector correspond to λ then λ^k is an eigen value of A^k and X is an eigen vector.
- ✓ If λ is an eigen value of matrix A then $\lambda \pm k$ is an eigen value of $A \pm kI$.
- ✓ If λ is an eigen value of matrix A then $k\lambda$ is an eigen value of kA .

METHOD – 5: EIGEN VALUES, EIGEN VECTORS AND EIGEN SPACE

C	1	Define the eigen value, eigen vector and eigen space.	
C	2	Let $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$ then find the eigen values of A , A^{25} , $3A$, A^{-1} , A^T , $A + 2I$ & $A^3 - 5I$. Answer: $-1, 3$ $-1, 3^{25}$ $-3, 9$ $-1, 1/3$ $-1, 3$ $1, 5$ & $-6, 22$.	
H	3	Find the eigen values of A and A^{-1} , where $A = \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}$. Answer: $1, 4$ & $1, 1/4$.	W-18 (2)

C	4	Find the eigen values of $A = \begin{bmatrix} -5 & 4 & 34 \\ 0 & 0 & 4 \\ 0 & 0 & 4 \end{bmatrix}$. Is A invertible? Answer: $-5, 0, 4$ & no.	
H	5	If $A = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 0 & 5 \\ 0 & 0 & 9 \end{bmatrix}$, then find the eigen values of the matrix A^T . Is A an invertible? Answer: $0, 1, 9$ & no.	
H	6	If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$ then find the eigen values of A^T and $5A$. Answer: $1, 2, 2$ & $5, 10, 10$.	
H	7	Find the eigen values of $A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ 12 & 15 & 3 \end{bmatrix}$ and hence deduce eigen values of A^5 and A^{-1} . Answer: $1, 2, 3$ & $1, 2^5, 3^5$ & $1, 1/2, 1/3$.	
H	8	Find the eigen values of A^9 for $A = \begin{bmatrix} 1 & 3 & 7 & 11 \\ 0 & 0.5 & 3 & 8 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}$. Answer: $1, 0.5^9, 0, 2^9$.	
T	9	Find the eigen values of the following matrices: (1). $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ (2). $\begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix}$ (3). $\begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$ (4). $\begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$ (5). $\begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$ (6). $\begin{bmatrix} 3 & 0 & -5 \\ 1/5 & -1 & 0 \\ 1 & 1 & -2 \end{bmatrix}$. Answer: (1). $1, 2$ (2). $2, 6$ (3). $4, 4$ (4). $1, 2, 2$ (5). $1, 2, 3$ (6). $0, -\sqrt{2}, \sqrt{2}$.	
C	10	Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then find the eigen values, eigen vectors corresponding to each eigen values of A. Also write the eigen vector space for each eigen values. Answer: 1 & -1 , $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ & $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $E_1 = \{ t \begin{bmatrix} 1 \\ 1 \end{bmatrix} / t \in \mathbb{R} \}$, $E_{-1} = \{ t \begin{bmatrix} -1 \\ 1 \end{bmatrix} / t \in \mathbb{R} \}$.	

H	11	Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$. Answer: $6, -1$ & $(1, 1)^T, (-2/5, 1)^T$.	
T	12	Show that if $0 < \theta < \pi$, then $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ has no real eigen values and consequently no eigen vectors.	
C	13	Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$. Answer: $2, 3, 5$ & $(-1, 1, 0)^T, (1, 0, 0)^T, (3, 2, 1)^T$.	
H	14	Find the eigen values and basis for eigen space for the matrix $A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$. Answer: $1, 3, 4$ & $\{(1, 2, 1)^T, (1, -1, 1)^T, (-1, 0, 1)^T\}$.	
T	15	Let $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$, then find eigen values, eigen vectors corresponding to each eigen values of A & write eigen space for each eigen values. Answer: $1, 2, 3$ & $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 1/2 \\ 1 \end{bmatrix}$ & $E_1 = \{t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} / t \in \mathbb{R}\}$, $E_2 = \{t \begin{bmatrix} -1 \\ 1/2 \\ 1 \end{bmatrix} / t \in \mathbb{R}\}$, $E_3 = \{t \begin{bmatrix} -1/2 \\ 1/2 \\ 1 \end{bmatrix} / t \in \mathbb{R}\}$.	
C	16	Find the eigen values and bases for the eigen space of the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Answer: $-1, -1, 2$ & $\{(-1, 1, 0)^T, (-1, 0, 1)^T, (1, 1, 1)^T\}$.	
H	17	Find the eigen values and corresponding eigen vectors for the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$. Answer: $1, 2, 2$ & $(-1, -1, 1)^T, (2, 1, 0)^T$.	W-18 (7)

H	18	Find the eigen values and bases for the eigen space of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{bmatrix}$ Answer: 1, 2, 2 & $\{(1, -1, 8)^T, (0, 0, 1)^T\}$.	
T	19	Find the eigen values and eigen vectors of $A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{bmatrix}$. Answer: 1, 1, 2 & $(-3, 1, 1)^T, (1, 0, 0)^T$.	
C	20	Find the eigen values and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{bmatrix}$ Answer: 2, 2, 2 & $(-1/3, -1/3, 1)^T$.	
H	21	Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$. Answer: 1, 1, 1 & $(1, 1, 1)^T$.	W-19 (4)

❖ ALGEBRAIC AND GEOMETRIC MULTIPLICITY OF AN EIGENVALUE:

- ✓ If eigenvalue of matrix A is repeated n-times, then n is called the Algebraic Multiplicity (A.M.) of λ .

Example: If $\lambda = 2, 2, 3$ then A.M. of $\lambda = 2$ is 2 and A.M. of $\lambda = 3$ is 1.

- ✓ The number of linearly independent eigenvectors corresponding to λ is called the Geometric Multiplicity (G.M.) of λ . i.e., $G.M. = \text{No. of variables} - \text{Rank}(A - \lambda I)$.
- ✓ $G.M. \leq A.M.$ for each eigen value.
- ✓ Procedure for finding A.M. and G.M. is as follow:
 - (1). Find eigen values of given matrix.
 - (2). Find A.M. using definition.
 - (3). Find eigen vectors for each eigen values.
 - (4). Find G.M. using definition.

METHOD – 6: ALGEBRAIC AND GEOMETRIC MULTIPLICITY

C	1	Explain the algebraic multiplicity and geometric multiplicity with example.	
H	2	Find the geometric and algebraic multiplicity of each eigen values of $A = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$. Answer: A. M. of 2 = 2 & G. M. of 2 = 1.	
C	3	Find the algebraic and geometric multiplicity of each of eigen value of $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Answer: A. M. (-1 & 2) = 2 & 1 & G. M. (-1 & 2) = 2, 1.	
C	4	Find A. M. and G. M. for each eigen values of $A = \begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$. Answer: A. M. (1) = 3 & G. M. (1) = 1.	
H	5	Find the eigen values and eigen vectors of $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$. Also find algebraic and geometric multiplicity of each eigen values. Answer: 5, 1, 1, $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, A. M. = 1, 2 & G. M. = 1, 2.	
T	6	Find the eigen values and eigen vectors of $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$. Also find algebraic and geometric multiplicity of each eigen values. Answer: 0, 3, 15, $\begin{bmatrix} 1/2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1/2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$, A. M. = 1, 1, 1 & G. M. = 1, 1, 1.	

❖ DIAGONALIZATION:

- ✓ A matrix A is said to be diagonalizable if it is similar to diagonal matrix.
- ✓ It means a matrix A is diagonalizable if there exists an invertible matrix P such that $P^{-1}AP = D$, where D is a diagonal matrix known as spectral matrix and P is known as modal matrix.

- ✓ An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors.
- ✓ If the eigenvalues of an $n \times n$ matrix are all distinct then the matrix is always diagonalizable.
- ✓ The necessary and sufficient condition for a square matrix to be diagonalizable is that the geometric multiplicity of each eigenvalue is same as its algebraic multiplicity.
- ✓ Procedure to check whether the given matrix is diagonalizable or not:
 - (1). Find the eigen values of a given matrix.
 - (2). If all the eigen values are distinct then the given matrix is diagonalizable, otherwise follow step (3).
 - (3). Find A.M. and G.M. for each eigen value.
 - (4). If both the multiplicities are same for all eigen value then the given matrix is diagonalizable.
- ✓ Procedure for finding a matrix P and $P^{-1}AP$ is as follow:
 - (1). Find the eigen values and eigen vectors of a given matrix.
 - (2). Construct matrix P by writing all the eigen vectors as its column vectors.
 - (3). Find P^{-1} .
 - (4). Find $P^{-1}AP$ and verify that $P^{-1}AP$ is diagonal matrix or not. The diagonal matrix $D = P^{-1}AP$ will have the eigenvalues on its main diagonal.
- ✓ Note that the order of the eigen vectors used to form P will determine the order in which eigen values appear on the main diagonal of D .
- ✓ $P^{-1}AP = D \Rightarrow A = PDP^{-1}$.
- ✓ Let A be a diagonalizable matrix and P is an invertible matrix such that $A = PDP^{-1}$ then $A^n = PD^nP^{-1}, \forall n \in \mathbb{N}$.
- ✓ If $D = \text{Diag}[a \ b \ c]$ then $D^n = \text{Diag}[a^n \ b^n \ c^n]$.
- ✓ Every diagonal matrix is diagonalizable.

METHOD – 7: DIAGONALIZATION

C	1	Check whether the following matrices are diagonalizable or not: $\begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$, $\begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 2 \end{bmatrix}$ & $\begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$. Answer: No, yes, no & yes.	
H	2	Check whether the following matrices are diagonalizable or not: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$ & $\begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$. Answer: Yes, yes, no & yes.	
T	3	Check whether the following matrices are diagonalizable or not: $\begin{bmatrix} 3 & 0 & -5 \\ 1/5 & -1 & 0 \\ 1 & 1 & -2 \end{bmatrix}$, $\begin{bmatrix} -2 & 0 & 1 \\ -6 & -2 & 0 \\ 19 & 5 & -4 \end{bmatrix}$, $\begin{bmatrix} -1 & 0 & 1 \\ -1 & 3 & 0 \\ -4 & 13 & -1 \end{bmatrix}$ & $\begin{bmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{bmatrix}$. Answer: Yes, yes, yes & no.	
T	4	Check whether the following matrices are diagonalizable or not: $\begin{bmatrix} 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ & $\begin{bmatrix} -1/3 & 0 & 0 & 0 \\ 0 & -1/3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1/2 \end{bmatrix}$. Answer: Yes & yes.	
T	5	If $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ then show that A is diagonalizable but B is not diagonalizable.	
C	6	Let $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal matrix. Answer: $P = \begin{bmatrix} 0 & 1/2 \\ 1 & 1 \end{bmatrix}$.	
H	7	Find the matrix P that diagonalize $A = \begin{bmatrix} 1 & 0 \\ 6 & -1 \end{bmatrix}$. Also determine $P^{-1}AP$. Answer: $P = \begin{bmatrix} 1/3 & 0 \\ 1 & 1 \end{bmatrix}$ & $P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.	

T	8	For the matrix $A = \begin{bmatrix} -4 & -6 \\ 3 & 5 \end{bmatrix}$, find the nonsingular matrix P and the diagonal matrix D such that $D = P^{-1}AP$. Answer: $P = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$ & $D = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$.	
C	9	Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Find an orthonormal matrix P such that $P^{-1}AP$ is diagonal matrix. Answer: $P = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$.	
H	10	Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. Find an orthogonal matrix P such that $P^{-1}AP$ is diagonal matrix. Answer: $P = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$.	
H	11	Find a matrix P that diagonalize $A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. Also determine $P^{-1}AP$. Answer: $P = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ & $P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.	
T	12	Find a matrix P that diagonalize $A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$. Also determine $P^{-1}AP$. Answer: $P = \begin{bmatrix} 1 & 2/3 & 1/4 \\ 1 & 1 & 3/4 \\ 1 & 1 & 1 \end{bmatrix}$ & $P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.	
C	13	Prove that $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$ is diagonalizable and use it to find A^{13} . Answer: $P = \begin{bmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ & $A^{13} = \begin{bmatrix} -8190 & 0 & -16382 \\ 8191 & 8192 & 8191 \\ 8191 & 0 & 16383 \end{bmatrix}$.	S-19 (7)

T	14	Let $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$. Find matrix P such that $P^{-1}AP$ is diagonal matrix and find A^5. Answer: $P = \begin{bmatrix} -1/2 & 1/2 & 2 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$ & $A^5 = \begin{bmatrix} 21856 & -10912 & 10912 \\ -10912 & 5488 & -5456 \\ 10912 & -5456 & 5488 \end{bmatrix}$.	
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❖ THE CAYLEY-HAMILTON THEOREM:

- ✓ Every square matrix satisfies its own characteristic equation. i.e., if A is an $n \times n$ matrix, and $P_A(\lambda) = 0$ is its characteristic equation, then $P_A(A) = \mathbf{O}_n$.
- ✓ That means the characteristic equation is satisfied by A.
- ✓ By using the Cayley-Hamilton theorem we can find A^{-1} and integer power of A. i.e., A^n .
- ✓ Procedure to verify Cayley-Hamilton theorem for 2×2 matrix is as follow:
 - (1). Find the characteristic equation of given matrix A.
 - (2). Put $\lambda = A$ in the characteristic equation and give it equation (1).
 - (3). Find A^2 using given matrix.
 - (4). Put the values of A^2, A, I in equation (1) and verify it.
 - (5). If you want to find A^{-1} using Cayley-Hamilton theorem then multiply equation (1) both sides by A^{-1} and substitute the values of A and I.
- ✓ Procedure to verify Cayley-Hamilton theorem for 3×3 matrix is as follow:
 - (1). Find the characteristic equation of given matrix A.
 - (2). Put $\lambda = A$ in the characteristic equation and give it equation (1).
 - (3). Find A^3 & A^2 using given matrix.
 - (4). Put the values of A^3, A^2, A, I in equation (1) and verify it.
 - (5). If you want to find A^{-1} using Cayley-Hamilton theorem then multiply equation (1) both sides by A^{-1} and substitute the values of A^2, A and I. By the similar way we can find A^{-2}, A^2, A^4 etc ...

METHOD – 8: THE CAYLEY - HAMILTON THEOREM

C	1	State Cayley-Hamilton theorem.	W-18 (1)
C	2	Using Cayley-Hamilton theorem find A^{-1} for $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$. Answer: $\frac{1}{5} \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$.	W-19 (4)
T	3	Using Cayley-Hamilton theorem, find A^{-1} , A^{-2} for $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$. Answer: $A^{-1} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ & $A^{-2} = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$.	
C	4	Verify Cayley-Hamilton theorem and find inverse for $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. Answer: $\frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$.	
H	5	State Cayley-Hamilton theorem and verify it for the matrix $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$.	S-19 (7)
T	6	Using Cayley-Hamilton theorem, find inverse of $\begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$. Answer: $\frac{1}{35} \begin{bmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ 6 & 1 & -10 \end{bmatrix}$.	
C	7	Verify Cayley-Hamilton theorem for $A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$ and hence find A^4 . Answer: $\begin{bmatrix} 193 & 160 & 144 \\ 224 & 177 & 160 \\ 272 & 224 & 193 \end{bmatrix}$.	
H	8	Verify Cayley-Hamilton theorem for $A = \begin{bmatrix} 6 & -1 & 1 \\ -2 & 5 & -1 \\ 2 & 1 & 7 \end{bmatrix}$ and hence find A^4 . Answer: $\begin{bmatrix} 2176 & -520 & 1400 \\ -1920 & 776 & -1400 \\ 1920 & 520 & 2696 \end{bmatrix}$.	

C	9	Determine A^{-1} by using Cayley-Hamilton theorem for matrix $A =$ $\begin{bmatrix} 1 & 3 & 2 \\ 0 & -1 & 4 \\ -2 & 1 & 5 \end{bmatrix}.$ Hence find the matrix represented by $A^8 - 5A^7 - A^6 + 37A^5 + A^4 - 5A^3 - 3A^2 + 41A + 3I$. Answer: $A^{-1} = \frac{1}{37} \begin{bmatrix} 9 & 13 & -14 \\ 8 & -9 & 4 \\ 2 & 7 & 1 \end{bmatrix}$ & $\begin{bmatrix} 13 & 8 & -40 \\ 16 & -11 & -16 \\ 16 & 8 & -27 \end{bmatrix}.$	
T	10	Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$. Answer: $\begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}.$	

